APPLIED PHYSICS LAB

I Semester: CSE, IT, CSE-Data Science, CSE-Cyber Security, CSE-AIML, CSIT, ECE, MECH, AERO, EEE

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS08	BSC	L	T	P	C	CIA	SEE	Total
		0	0	3	1.5	40	60	100

Course Objectives:

The course should enable the students to:
1. Understand the temperature and other dependent properties of semiconductor devices and their usages in different applications.
2. To know the Laser and Fibre optic technologies and its applications in real time scenario.
3. Understand the electromagnetic properties using experimental knowledge.
4. Understand the method of least squares fitting.

LIST OF EXPERIMENTS

Experiment-1	**ENERGY GAP OF P-N JUNCION DIODE:** To determine the energy gap of a given semiconductor diode
Experiment-2	**R CELL:** To study the V-I and V-P characteristics and determine the fill factor of solar cell
Experiment-3	**LIGHT EMITTING DIODE:** To study the characteristics of LED by plotting V-I graph and determine the threshold value of given LEDs
Experiment-4	**HALL EFFECT:** To determine Hall co-efficient and charge carrier concentration of a given semiconductor
Experiment-5	**PIN PHOTO DIODE:** To study the V-I Characteristics of Photo Diode with respect to intensity of light
Experiment-6	**OPTICAL FIBRE:** To determine the numerical aperture and acceptance angle of an optical fiber
Experiment-7	**LASER:** To determine the wavelength of a given laser source by using diffraction grating method
Experiment-8	**NEWTON'S RINGS:** To determine the radius of curvature of a given Plano convex lens by forming Newton's rings
Experiment-9	**THERMISTOR:** To study the variation of resistance with respect to temperature using thermistor
Experiment-10	Understanding the method of least squares - Torsional pendulum as an example
Experiment-11	**PLANCK'S CONSTANT:** To determine value of planck's constant by measuring radiation in fixed spectral range
Experiment-12	**STEWART GEE'S EXPERIMENT:** To study the variation of magnetic field along the axis of a circular coil and calculation of magnetic flux

Note: Students have to perform any 8 experiments

Reference Books:

1. "Applied Physics Lab Manual"- Dr. Radhika Devi, Mr. A V Laxman Rao, N. Noel
2. S. Balasubramanian, M.N Srinivasan "A Text book of practical Physics" – S Chand Publishers, 2017.

Course Outcomes:
Students should be able to:

1. Analyze the electric properties of semiconductor materials by determining energy gap of semiconductors, charge carrier concentration in Semiconductors using Hall effect and threshold voltage of LEDs, photo current in Photo diodes, solar cell, and temperature effect on resistance using thermistor.

2. Identify the optical properties of light such as diffraction phenomenon using grating material for calculation of the wavelength of Laser and acceptance angle, NA of optical fiber using OFC and determine the value of Plank's constant using a light source and interference by using Newton's rings.

3. Analyze the electromagnetic properties of a current carrying coil by using Stewart Gee's experiment.

4. Analyze the least squares fitting method for data analysis using experimental data of Tensional Pendulum.